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# PDS background activity and dead time

Current estimate, simulation chain, and remaining uncertainty

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# Candidate activity is not board acceptance

NOMINAL INTRINSIC-LAr RESULT

A VD cathode channel  
sees 3.59 times the HD  
candidate activity.

The mixture is dominated by atmospheric  
 $^{39}\text{Ar}$ ; at the same 0.7-PE threshold:  
87.47 kHz/ch for VD cathode and 24.34 kHz/ch  
for HD.

## 01 LArSoft output

Candidate times and rates.

## 02 Firmware estimate

Channel-local busy gate.

## 03 Not yet modeled

Shared board transport.

# Assumptions: source, threshold, and pulse shape

dunesw v10\_22\_00d01 | LArSoft v10\_22\_00 | art 3.14.04

| Item              | Condition used for the headline result                                                                                                                                                       |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source            | Uniform intrinsic-LAr mixture: atmospheric $^{39}\text{Ar}$ at 0.964 Bq/kg, plus released $^{85}\text{Kr}$ , $^{42}\text{Ar}$ , and $^{42}\text{K}$ ; eight independent shards per detector. |
| Optical threshold | Every local candidate above 0.7 PE is counted. <b>There is no 10-MeV trigger cut.</b> Energy is retained only for classification and auditing.                                               |
| Pulse model       | The 16 Waffles NP02 cathode C-channel large_pulses_zero_subtract waveforms form one common SPE-shape set for HD and VD.                                                                      |

## MODEL ASSUMPTION

### The common C-channel pulse set isolates geometry and activity

It is not evidence that HD and VD have identical gain, noise, PDE, or technology-specific SPE shapes.

# What we simulated—in chronological order

| Step | Operation            | Meaning                                                                                                              |
|------|----------------------|----------------------------------------------------------------------------------------------------------------------|
| 1    | Physical source      | LArSoft draws radioactive decays during a known time interval, or one gamma from a controlled energy/face cell.      |
| 2    | Full geometry        | Geant4 transports particles through the complete HD-v6 or VD-v7 detector geometry.                                   |
| 3    | Optical response     | Energy deposited in liquid argon produces light and the detector-specific photon response.                           |
| 4    | Common pulse model   | Detected light is shaped with the same 16 Waffles NP02 cathode-C templates in both technologies.                     |
| 5    | Candidate + firmware | A strict $> 0.7$ -PE crossing forms a candidate; XC fixed-record busy time is applied afterward, channel by channel. |

**Physical windows give Hz; stratified particles give response.**

A response matrix becomes a rate only after applying the physical source weights.

# Full geometry and channel denominator

| Item                | Condition used for the headline result                                                                                   |
|---------------------|--------------------------------------------------------------------------------------------------------------------------|
| FD-HD full detector | LArSoft dune10kt_v6_full: 6,000 APA-integrated optical hardware channels; HD is the reference.                           |
| FD-VD full detector | LArSoft dunevd10kt_3view_30deg_v7_full: 1,344 channels—640 cathode, 640 long-wall, and 64 short-wall.                    |
| Optical response    | Geometry and photon transport remain technology-specific. Only the Waffles C-channel SPE pulse-shape set is common.      |
| Denominator         | All hardware channels in the relevant population, including quiet channels; units are candidates per channel per second. |

## DEFINITION

### **Candidate** $\neq$ **accepted frame**

A candidate is a threshold-crossing opportunity on one channel. Firmware acceptance is evaluated afterward.

# What drives the nominal intrinsic-LAr rate

| Component                          | Generator content                                      | Share of generated primaries |
|------------------------------------|--------------------------------------------------------|------------------------------|
| $^{39}\text{Ar}$                   | Atmospheric argon at 0.964 Bq/kg                       | 89.4%                        |
| $^{85}\text{Kr}$                   | Released intrinsic-LAr normalization                   | 10.6%                        |
| $^{42}\text{Ar}$ , $^{42}\text{K}$ | Released intrinsic-LAr normalizations                  | < 0.02% combined             |
| Other families                     | Detector materials, radon chains, and external sources | Not in this baseline         |

## CORRECTION

### The 24.34/87.47-kHz numbers are not Ar-39-only

The source labels are merged before optical simulation. We can state the generated-primary mix, but cannot assign the final candidates to one isotope from this sample.

# Background scope: what is not closed

| Family              | Current content                                                                        | Status                                            |
|---------------------|----------------------------------------------------------------------------------------|---------------------------------------------------|
| External $\gamma$   | Cavern-wall and foam; cryostat- and cavern-neutron-induced $\gamma$ generators         | One physical window per family; <b>pilot only</b> |
| Neutrons / captures | Actual cavern neutrons, capture timing and secondary $\gamma$ , alpha-capture $\gamma$ | Not closed; excluded                              |
| Remaining radon     | $^{214}\text{Bi}$ and the missing $^{219}\text{Rn}/^{220}\text{Rn}$ chain rows         | Incomplete; excluded                              |
| Electronics         | Dark counts, line/electronics noise, afterpulsing, and cross-talk                      | No validated physical-rate component              |

## NO TOTAL-BACKGROUND CLAIM

### The gamma extension is a first physical-window estimate

Its candidate ticks are overlaid with the nominal stream before XC busy time. More gamma windows and the missing families are still required.

# The matched gamma-response pilot is complete

Conditional detector response—physical gamma weights are not yet applied

| Detector | Incident $\gamma$ | Active-LAr deposit | > 0.7-PE response | Probability |
|----------|-------------------|--------------------|-------------------|-------------|
| FD-HD    | 10,000            | 2,458              | 2,358             | 23.58%      |
| FD-VD    | 10,000            | 1,868              | 1,782             | 17.82%      |

## EVENT RESPONSE

$$\text{VD} / \text{HD} = 0.756$$

This compares the same artificial proposal: ten energy bins, five entry faces, and 200 independent gammas per cell.

## CHANNEL-NORMALIZED YIELD

$$\text{VD} / \text{HD} = 0.912$$

Candidate counts divided by 1,344 VD or 6,000 HD channels and by the same number of injected gammas.

HD low-z gave zero active-LAr deposits despite verified inward directions; retain as a pilot observation and validate more densely.

# What is—and is not—a background-rate result

| Quantity                           | Status         | Correct interpretation                                                                                                       |
|------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------|
| Nominal intrinsic LAr              | Rate-ready     | $^{39}\text{Ar}$ at 0.964 Bq/kg plus $^{85}\text{Kr}$ , $^{42}\text{Ar}$ , and $^{42}\text{K}$ ; eight windows per detector. |
| External- $\gamma$ response matrix | Complete       | Conditional efficiency versus energy and face; not Hz/channel.                                                               |
| Physical external- $\gamma$ rate   | Pilot complete | Released physical generators in full geometry; one window per source family.                                                 |
| Total background stream            | Not closed     | Requires all source families overlaid in time before XC and shared-board back-pressure.                                      |

Two labels are kept separate: nominal intrinsic LAr and intrinsic LAr + gamma pilot.  
Neither is yet a complete board-level background model.

# Physical gamma pilot: first absolute rates

One full-geometry physical window per source family; strict  $> 0.7$  PE

| Population        | Cavern wall<br>kHz per channel | Foam  | Total gamma |
|-------------------|--------------------------------|-------|-------------|
| HD APA-integrated | 18.805                         | 0.719 | 19.523      |
| VD cathode        | 4.731                          | 0.184 | 4.915       |
| VD long wall      | 21.805                         | 0.943 | 22.749      |
| VD short wall     | 15.461                         | 0.675 | 16.136      |

## PILOT LIMIT

### Zero observed is not zero expected

The cryostat-neutron- and cavern-neutron-gamma generators produced one primary each and no candidates in these windows. The deposited-energy audit also timed out, so the energy population is not yet validated.

# Nominal intrinsic-LAr activity depends on position

| Population        | Mean<br>kHz per channel | Min-max      | Mean / HD |
|-------------------|-------------------------|--------------|-----------|
| HD APA-integrated | 24.34                   | 6.68-44.77   | 1.00      |
| VD cathode        | 87.47                   | 58.99-107.03 | 3.59      |
| VD long wall      | 30.49                   | 20.03-40.16  | 1.25      |
| VD short wall     | 22.05                   | 16.21-27.69  | 0.91      |

READ THE RANGE

**Mean  $\neq$  maximum**

Hottest VD cathode  
channel: **107.03** kHz/ch.

NOMINAL RATIO

**Cathode / HD = 3.59**

87.47/24.34, before  
firmware loss.

# Candidate-level gamma overlay: provisional load

shards

| Population        | Intrinsic LAr<br>candidate | Gamma<br>kHz/ch | Combined      | Accepted [kHz/ch] |               |
|-------------------|----------------------------|-----------------|---------------|-------------------|---------------|
|                   |                            |                 |               | 1024              | 512           |
| HD APA-integrated | 24.338                     | 19.523          | <b>43.855</b> | 21.030            | <b>26.152</b> |
| VD cathode        | 87.467                     | 4.924           | <b>92.383</b> | 34.534            | <b>48.297</b> |
| VD long wall      | 30.490                     | 22.794          | <b>53.273</b> | 25.020            | <b>32.167</b> |
| VD short wall     | 22.050                     | 16.136          | <b>38.181</b> | 21.264            | <b>26.238</b> |

## WHAT THE OVERLAY ADDS

### Cross-source candidate-level pileup

Busy rejection is computed after merging timestamps: HD 52.05% → 40.37%, and VD cathode 62.62% → 47.72%, for 1024 → 512 samples.

VD gamma candidates are clipped from the native 8.57-ms window to the nominal 8.50-ms window before overlay. Analog pileup and shared board back-pressure are not modeled.

The conversion to offered board traffic is explicit

$$W_{\text{rec}} = 8 + 7 \left\lceil \frac{N_s}{32} \right\rceil, \quad B_{\text{rec}} = 64W_{\text{rec}}, \quad T_{\text{board}} = N_{\text{ch}} R_{\text{cand/ch}} B_{\text{rec}}$$

| Case            | Channels | Rate/ch   | Bits/record | Offered traffic |
|-----------------|----------|-----------|-------------|-----------------|
| VD cathode, 512 | 32       | 87.47 kHz | 7,680       | 21.5 Gbit/s     |
| HD, 512         | 40       | 24.34 kHz | 7,680       | 7.48 Gbit/s     |

512  $\rightarrow$  120 64-bit words; 1024  $\rightarrow$  232 words = 14,848 bits.

#### CHANNEL COUNT MATTERS

**Homogeneous VD-cathode / HD board load = 2.87**

This is an offered-load illustration, not an accepted-output prediction or a detector wiring claim.

# Why $14 \times 512 \times 32$ is not yet throughput

**BITS IN ONE BOARD-WIDE SET**

**A size, not a rate**

$14 \text{ bit/sample} \times 512 \text{ sample/frame} \times$   
 $32 \text{ ch} = 229,376 \text{ bit}.$

**BITS PER SECOND**

**Multiply by activity**

With the modeled record overhead:  
 $32 \text{ ch} \times R_{\text{ch}} \times 7,680 \text{ bit/frame}.$

**512 + 50% OVERLAP**

**Overlap changes acceptance opportunities, not record size**

Each fragment is still 7,680 bits. The 256 overlapping samples are retransmitted unless firmware coalesces them:  $T = R_{\text{accepted,overlap}} \times 7,680.$

# What the present firmware replay actually does

| Stage               | Included here                                         | Still separate                                                                |
|---------------------|-------------------------------------------------------|-------------------------------------------------------------------------------|
| Candidate formation | LArSoft timestamp for each strict $> 0.7$ -PE episode | Continuous ADC overlay across source families                                 |
| XC channel logic    | Fixed busy gate: 1,037 cycles for 1024; 525 for 512   | ADC-level XC FIR/CFD decision and efficiency                                  |
| Window overlap      | Record size is unchanged                              | 512 + overlap acceptance/coalescing policy                                    |
| Shared transport    | None                                                  | Board map, FIFO, lanes, serializer, Hermes ready, and 10-Gbit/s back-pressure |

**Keep the evidence separate:** the  $> 89\%$  result is from a different XC injected-signal test at 1 PE and above; it is not an efficiency measured by this replay.

# Hermes back-pressure changes acceptance

Intrinsic-LAr + one-window gamma pilot; homogeneous-board illustration

Sender input:  $T_{\text{in}} = R_{\text{accepted, local}} B_{\text{record}}$ ; queue growth:  $dQ/dt = \max(0, T_{\text{in}} - C_{\text{payload}})$ , where  $C_{\text{payload}} < 10 \text{ Gbit/s}$ .

| Population    | 512 demand $\rightarrow T_{\text{in}}$ | 1024 demand $\rightarrow T_{\text{in}}$ | Ideal-link test      |
|---------------|----------------------------------------|-----------------------------------------|----------------------|
| HD            | 13.47 $\rightarrow$ 8.03               | 26.05 $\rightarrow$ <b>12.49</b>        | 1024 saturates       |
| VD cathode    | 22.70 $\rightarrow$ <b>11.87</b>       | 43.90 $\rightarrow$ <b>16.41</b>        | <b>both saturate</b> |
| VD long wall  | 13.09 $\rightarrow$ 7.91               | 25.31 $\rightarrow$ <b>11.89</b>        | 1024 saturates       |
| VD short wall | 9.38 $\rightarrow$ 6.45                | 18.14 $\rightarrow$ <b>10.10</b>        | 1024 at limit        |

All traffic values are Gbit/s for one homogeneous board and one 10-Gbit/s sender. “Demand” is before the local gate;  $T_{\text{in}}$  is after it.

## CONSEQUENCE

**The ideal 10-Gbit/s comparison is already unfavorable**

All 1024-sample populations meet or exceed line rate; VD cathode also exceeds it at 512.

Usable payload is below 10 Gbit/s, so final loss requires Hermes ready, FIFO depth, and the real channel-to-sender map.

# 512 samples improves the channel-local estimate

Nominal intrinsic-LAr baseline; fixed-record busy gate applied independently per channel

| Population        | Accepted rate [kHz/ch] |              | Rejected while busy |              |
|-------------------|------------------------|--------------|---------------------|--------------|
|                   | 1024                   | 512          | 1024                | 512          |
| HD APA-integrated | 15.45                  | <b>17.91</b> | 36.5%               | <b>26.4%</b> |
| VD cathode        | 33.62                  | <b>46.53</b> | 61.6%               | <b>46.8%</b> |
| VD long wall      | 18.78                  | <b>22.27</b> | 38.4%               | <b>27.0%</b> |
| VD short wall     | 15.47                  | <b>17.76</b> | 29.9%               | <b>19.5%</b> |

## SUPPORTED

### The direction of the effect

Shortening 1024 to 512 samples increases accepted candidate rate in every population tested.

## NOT SUPPORTED YET

### A board-level dead-time number

This replay has no shared FIFO, serializer, lane assignment, or Hermes ready/back-pressure.

# The remaining uncertainty is architectural

## WHAT WE KNOW

### Detector-side activity

Nominal intrinsic-LAr rates and ranges, a one-window physical gamma pilot, VD populations, and the direction of the 512-sample effect.

## WHAT IS REQUIRED

### A board-aware replay

Map channel → board → sender; replay correlated times through FIFO, serializer, Hermes ready, and the 10-Gbit/s link.

## REPORTING RULE

### Keep the labels separate

Candidate activity, channel-local acceptance, and offered sender traffic are different quantities. Reserve *board throughput/dead time* for the mapped model.

# What we can say today

## DETECTOR ACTIVITY

### The VD cathode is demanding

Intrinsic LAr: 87.47 versus HD 24.34 kHz/ch (3.59 $\times$ ). Gamma pilot: 92.38 versus 43.85 (2.11 $\times$ ).

## RECORD LENGTH

### 512 samples helps

VD-cathode rejection: 61.6%  $\rightarrow$  46.8% in the baseline; 62.6%  $\rightarrow$  47.7% with the gamma pilot.

## TRANSPORT LIMIT

### One 10-Gbit/s link is tight

Gamma pilot after the local gate: VD cathode 11.87 Gbit/s at 512; 16.41 Gbit/s at 1024.

## NEXT CLOSURE

### The estimate remains provisional

Repeat gamma windows; add missing backgrounds; validate energy; then map boards, FIFO, lanes, serializer, and Hermes ready.